SOVEREIGNNEXUS LLC | EXECUTIVE PITCH DECK — SLIDE 1/10

Bare-Metal Deterministic Edge AI Substrate

The 1=1=1 Axiom

Human Intent ≡ Digital Code ≡ Bare-Metal Hardware Execution. Eliminating probabilistic variance across continuous edge loops.

Hardware Physics

Bit-exact Q16.16 integer math, SWHT spectral kernels, 105.0°C Metabolic Governor, and sub-15MB active memory footprint.

Federal Credentials

SAM.gov UEI: K5DALREZFGH6 | CAGE Code: 1AQG5 | NAICS: 541715 | Founder & Architect: David John Niedzwiecki Jr.

THE PROBLEM — SLIDE 2/10

Commercial AI Is Trapped in Non-Deterministic Drift & Context Decay

FP16/FP32 Mantissa Noise

Micro-scale floating-point rounding errors compound over execution loops into catastrophic trajectory drift and goal abandonment.

Superficial Soft Prompts

Post-hoc soft wrappers and guardrails fail because the underlying mathematical execution layer remains non-deterministic.

High-Stakes Risk

Safety-critical defense, edge robotics, and enterprise systems cannot afford cloud latency or probabilistic guesswork.

THE SOLUTION — SLIDE 3/10

The 1=1=1 Axiom of Functional Equivalence

Human Intent ≡ Logic ≡ Silicon

Binding human intent directly to digital code and physical hardware execution without probabilistic variance.

Bit-Exact Integer Math

Replaces floating-point MAC units with discrete integer arithmetic and deterministic matrix transformations.

Substrate Independence

Full air-gapped execution running directly on 8GB commodity edge hardware without cloud GPU dependencies.

TECHNICAL MOAT — SLIDE 4/10

1.58-Bit Ternary Engine & TQ1_0 10.09x Compression

Base-3 Galois Field GF(3)

Parameter weights strictly constrained to balanced ternary values $W \in \{-1, 0, +1\}$, carrying 1.58 bits/param.

TQ1_0 Byte Density

$3^5 = 243 \leq 256$ losslessly packs 5 trits into 1 byte, shrinking a 7B model footprint from 14.0 GB (FP16) down to 1.38 GB.

Zero-Copy mmap Paging

Kernel virtual memory address space paging off NVMe SSDs sustains a flat 0.00 KB active RAM heap delta.

COMPUTE EFFICIENCY — SLIDE 5/10

SWHT Integer Kernels & Zero-State Noise Filter

SWHT Spectral Additions

Sovereign Walsh-Hadamard Transform replaces floating-point MACs with 100% integer addition/subtraction passes.

71.4% Dynamic Energy Cut

Reduces dynamic compute energy to ~1.15 pJ/op and thermal drag by >85% compared to standard FP16 execution.

Zero-State Noise Quarantine

The neutral ternary state (0) quenches unverified probabilistic noise directly at the matrix boundary.

ZERO-TRUST SECURITY — SLIDE 6/10

Lens 09 Shannon Sentinel & 105.0°C Metabolic Governor

Lens 09 AirLock Sentinel

Calculates character payload entropy in <0.1 ms, instantly isolating high-entropy injections (H(X) > 4.8) and null-byte exploits.

EDCBA Verification Loop

Evaluate → Decompose → Converge → Bind → Anchor loop locks states to the E8 Root Lattice via SLERP (||W|| = 1.0000).

105.0°C Metabolic Governor

Monitors Linux /proc telemetry and triggers an un-overrideable Landauer limit power halt (k_B * T * ln 2) to stop bit-flips.

EMPIRICAL PROOFS — SLIDE 7/10

Sub-15MB Active Heap & 0.0% Trajectory Replay Drift

12 MB Active Heap Footprint

Empirically verified continuous execution running on a 12 MB active RAM footprint with flat 0.00 KB overhead delta.

0.0% Trajectory Replay Drift

500/500 unbroken execution passes satisfying DoD Directive 3000.09 V&V; standards for autonomous systems.

SHA-256 Merkle BlockDAG

Sequential state transitions sealed into append-only SQLite WAL ledgers (nexus_checkpoints.db) with ECDSA signatures.

DUAL-USE MARKET — SLIDE 8/10

Defense Autonomy & Enterprise Edge Security ($150B+ TAM)

Tactical Defense Autonomy

Un-hackable drone swarms, littoral sensor meshes, and USAF A-GRA Collaborative Combat Aircraft (CCA) integration.

Industrial IoT & Robotics

Embedded zero-heat microcontrollers operating in denied, disconnected, or contested electronic environments.

Enterprise B2B Security

Air-gapped enterprise memory nodes eliminating internal data leakage (AgentLeak) and context amnesia.

COMMERCIALIZATION — SLIDE 9/10

Three-Tiered B2B & Federal Revenue Engine

Per-Node Runtime Licensing

Annual per-device runtime licensing for edge hardware manufacturers, defense primes, and robotics OEMs.

Federal Defense Contracts

Direct SBIR/STTR Phase I/II proposals, USAF A-GRA subcontracts, and agency Statements of Work (SOW).

Custom TQ1_0 Quantization

Enterprise airlock staging, zero-trust schema integration, and specialized model quantization services.

THE ASK — SLIDE 10/10

Seed Capital to Scale Pilot Deployments & IP Protection

Seed Capital Target

Seeking seed investment to expand bare-metal testing suites, formalize commercial pilots, and file foundational utility patents.

12-Month Milestones

Complete USAF CCA pilot integration, deploy 3 enterprise B2B airlock SOWs, and scale C-FFI benchmark tooling.

Contact & Credentials

Founder & Architect: David John Niedzwiecki Jr. | SAM.gov UEI: K5DALREZFGH6 | CAGE: 1AQG5 | NAICS: 541715